

gautham.balajis@gmail.com
+91 9080302593
Chennai, India 600041

Technical Skills

Languages: Python, C++, Java, SQL, JavaScript

AI & Agentic Systems:

LangChain, Microsoft Autogen, RAG, LLM Agents, Tool Orchestration, Prompt Engineering

Backend & Full Stack: Flask, FastAPI, React, JWT Authentication

Tools & Platforms: Git, GitHub, GitHub Actions, Hugging Face Spaces

Education

B.Tech in Computer Science and Engineering (AI & ML) | 2027 | CGPA: 8.30
Vellore Institute of Technology, Chennai

Websites, Portfolios, Profiles

- [LinkedIn](#)
- [GitHub](#)
- [Portfolio](#)

GAUTHAM BALAJI

Summary

Software Engineer with experience building scalable backend systems, full-stack applications, and production-grade AI platforms. Skilled in designing distributed systems, developing enterprise software, and integrating intelligent workflows using modern AI technologies. Strong foundation in machine learning, system design, and cloud-native application development, with a focus on building reliable, performant, and maintainable software.

Experience

Data Science Intern

Sundaram Finance 05/2025 – 07/2025

- Formulated an end-to-end customer churn prediction pipeline using real-world banking data (10,000+ records) to support proactive retention strategies in an operational banking environment.
- Conducted extensive EDA, feature engineering, and outlier treatment to uncover churn drivers such as age, product count, and activity level.
- Trained and evaluated Logistic Regression, Random Forest, XGBoost, and Stacked Ensemble models using ROC AUC, Recall, and F1-score.
- Improved churn recall from 0.21 (baseline) to 0.73 using ensemble learning, prioritizing minority-class detection.
- Applied SHAP explainability to interpret individual predictions and global feature importance for business stakeholders.
- Deployed an interactive Streamlit application on Hugging Face Spaces with CI/CD automation via GitHub Actions.

Full Stack Developer

Admrls 08/2025 - Current

- Developed enterprise-grade full-stack applications spanning backend services, web interfaces, and AI-powered workflows, contributing to scalable software solutions across multiple client engagements.
- Co-developed a production-ready multi-agent AI platform built on Microsoft AutoGen, implementing modular agent orchestration, task routing, lifecycle management, and enterprise observability for reliable agent execution.
- Contributed to the architecture of scalable agent infrastructure by implementing structured logging, role-based access control (RBAC), execution tracing, and fault recovery mechanisms aligned with enterprise deployment standards.
- Co-developed the commercialization of an enterprise video conferencing platform by implementing multi-tenant architecture, secure recording management, administrative workflows, and platform capabilities required for scalable SaaS deployment.
- Collaborated across product, design, and engineering teams to deliver production-ready software, troubleshoot complex system issues, and improve scalability for customer-facing enterprise products.

Technical Specialist – CrftHQ

CrftHQ (student-led innovation accelerator empowering 30,000+ student creators with resources, mentorship, and venture support) 07/2025 - Current

- Designed and deployed **automation systems**, including streamlined **email communication workflows** to improve outreach and engagement.
- Collaborated with cross-functional teams to integrate **AI, design, and automation** into CrftHQ’s operations, enhancing scalability and impact.

Projects

Explainable AI Chess Engine

Python, TensorFlow, Scikit-Learn

- Developed a fully playable explainable AI chess engine combining a CNN-based position evaluator with classical chess heuristics to generate interpretable move recommendations.
- Designed a hybrid evaluation pipeline by integrating neural position scoring, heuristic features, and a Ridge regression model trained against Stockfish evaluations for principled move ranking.
- Optimized inference through batched evaluation and move reranking with one-ply lookahead, enabling responsive gameplay while improving decision quality.
- Built an interactive Flask web application featuring real-time gameplay, live analysis, top move recommendations, and human-readable explanations supported by Integrated Gradients visualizations.

VERA – Real-Time Voice AI Agent

Gemini Live, Twilio, RAG, Tool Orchestration

- Built a real-time AI voice agent integrating Gemini Live and Twilio APIs for conversational automation.
- Implemented multi-tool orchestration enabling dynamic tool selection during live calls.
- Integrated RAG pipeline with vector-based retrieval to ground responses in domain-specific knowledge.
- Optimized latency and response flow for near real-time conversational performance.
- Designed structured conversation memory handling for contextual continuity across sessions.

Legal NLP Knowledge Graph System

Python, spaCy, BERT, NetworkX

- Developed an end-to-end legal intelligence platform that extracts entities from judicial documents, constructs knowledge graphs, and enables intelligent legal case analysis.
- Engineered a hybrid retrieval pipeline combining semantic search with graph-based reasoning to improve similar-case discovery and explainable legal research.
- Built a modular NLP pipeline for legal entity extraction, relationship modeling, case classification, and interactive knowledge graph visualization using Streamlit.
- Optimized latency and response flow for near real-time conversational performance.
- Integrated SHAP and LIME explainability techniques to provide interpretable classification decisions and highlight influential legal concepts within judgments.